



Wavelet Denoising

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Overview :

1. Definition and Purpose.
2. Mathematical Foundation.
3. Implementation.
4. Advantages and Disadvantages.
5. Practical Example.

The Challenge of Noisy Signals

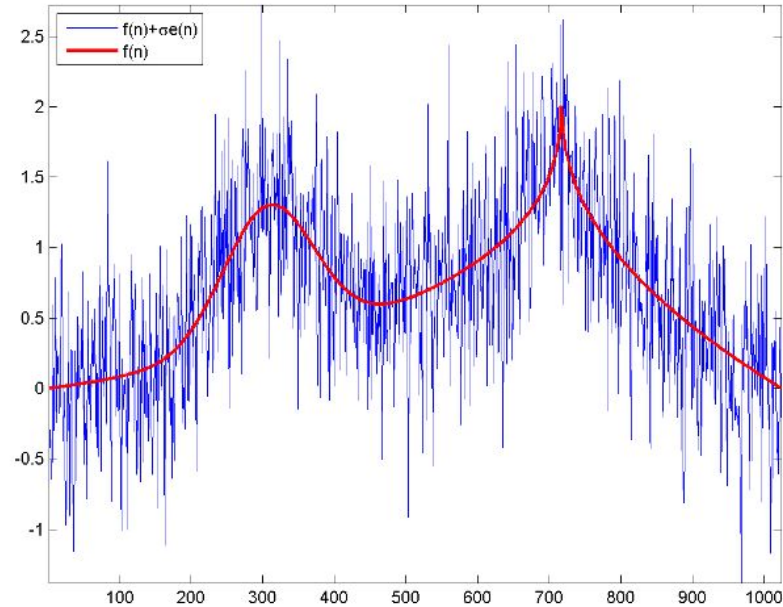


fig: 1 [1]

Noisy Signal $y(t)$ we observe is a combination of the True, Clean Signal $s(t)$ we actually want, plus unwanted Noise $n(t)$

$$y(t) = s(t) + n(t)$$

The Core Idea: Why Use Wavelets?

The Challenge: How can we separate a meaningful signal from random noise when they are mixed together?

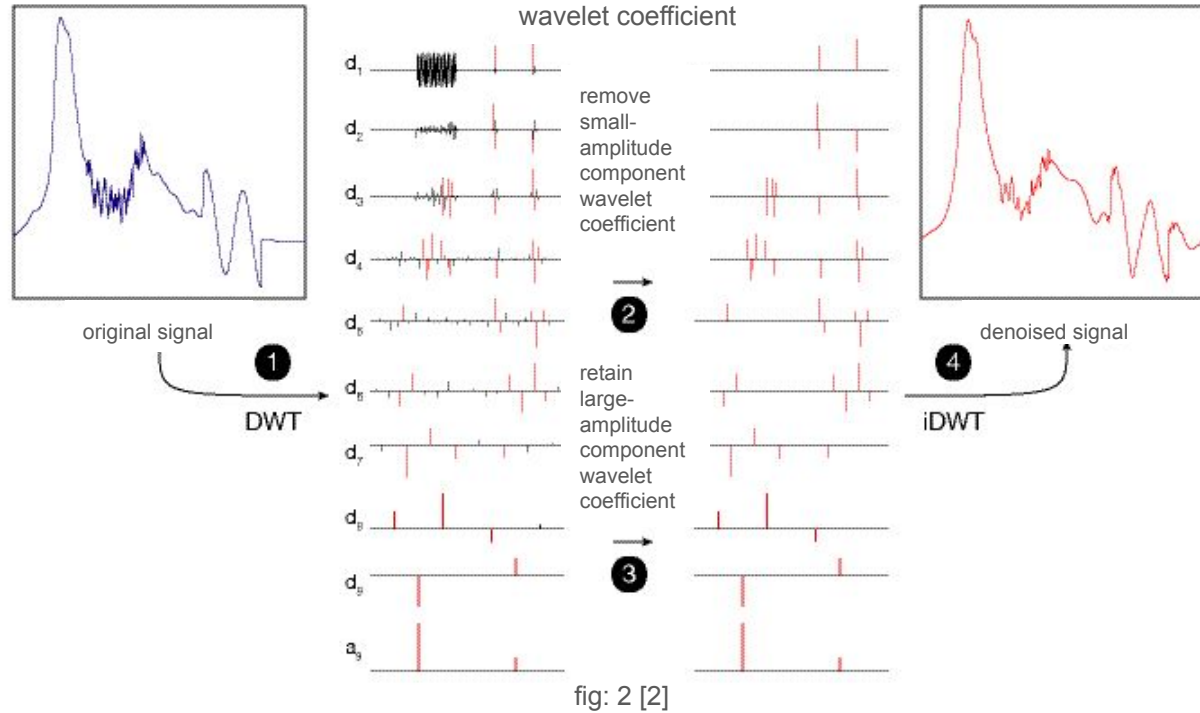
The Wavelet Solution: The Wavelet Transform is the key. It analyzes the signal and separates its components based on their scale.

This creates a crucial difference in how the signal and noise appear:

- **True Signal:** Its energy gets concentrated into a **few, large** wavelet coefficients. These represent the signal's important features.
- **Noise:** Its energy is spread thinly across **many, small** wavelet coefficients.

The Result: Because the signal and noise look so different in the wavelet domain, we can create a rule to discard the small coefficients (the noise) and keep the large ones (the signal).

Denoising Process



How to Implement Wavelet denoised filter

Following are the steps for implementing a wavelet denoised filter in real-world signals

Step 1: Choose a wavelet

Step 2: Decompose the Signal (Wavelet Transform)

Step 3: Apply Thresholding (Noise Reduction)

Step 4: Reconstruct the signal

Step 5: Evaluate Performance

Common Methods for Evaluation

1. Mean Squared Error (MSE)

$$\text{MSE} = (1/N) \times \sum (x_i - \hat{x}_i)^2 \text{ from } i = 1 \text{ to } N$$

2. Signal-to-Noise Ratio (SNR)

$$\text{SNR} = 10 \times \log_{10} [\sum(x_i^2) / \sum(x_i - \hat{x}_i)^2] \text{ for } i = 1 \text{ to } N$$

x = original clean signal

$\hat{x}_i$ = denoised signal

N = Length of signal

Wavelet-Based Denoising of a Noisy Sine Signal

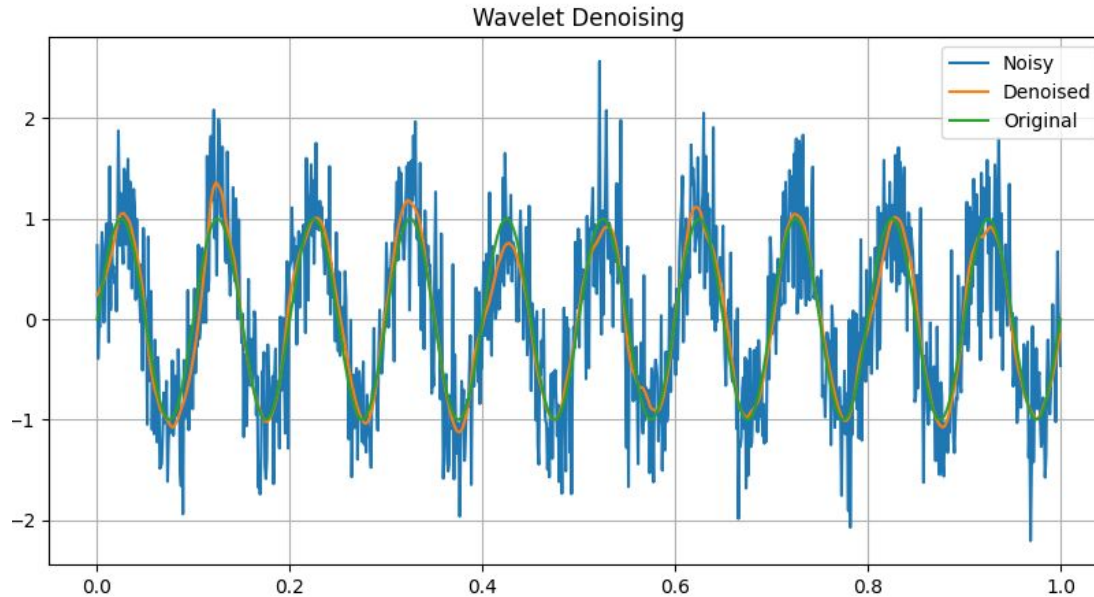


fig: 3

The code simulates a noisy sine wave and denoises it using a 4-level discrete wavelet transform with soft thresholding, then reconstructs and plots the original, noisy, and denoised signals for comparison.

Advantages and Disadvantages of Wavelet denoising Filter

Advantage:

- Time-frequency Localization
- Efficient noise suppression

Disadvantage:

- Choice of Wavelet and Threshold
- Non-optimal for colored noise

Practical Example in Seismic Signal

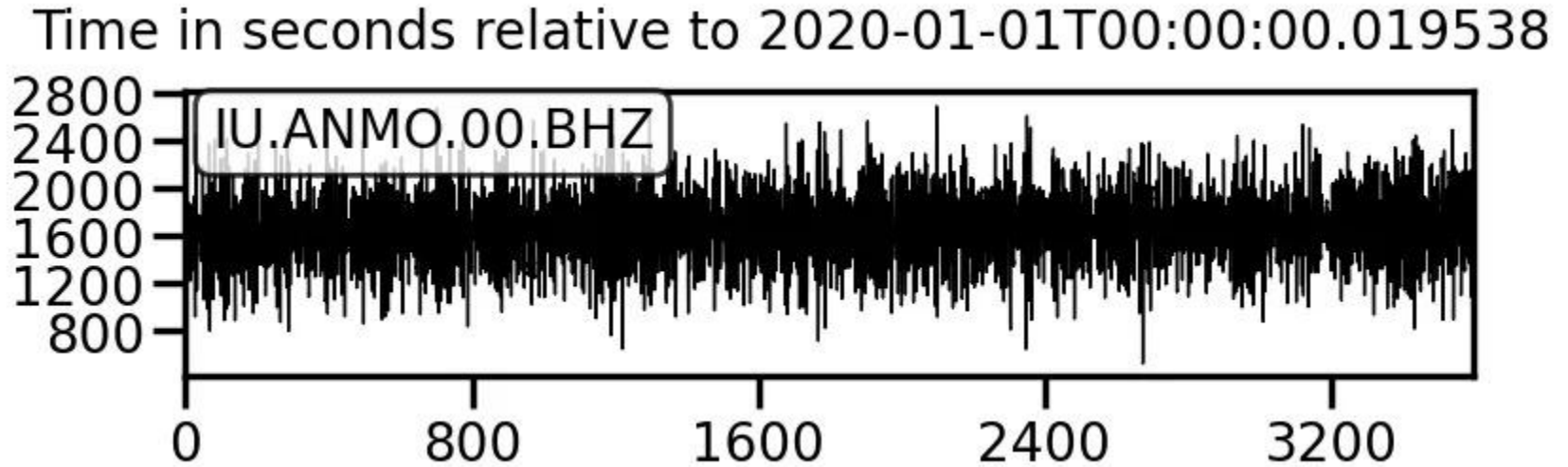


fig: 4 [3]

Before and After impact Wavelet denoising

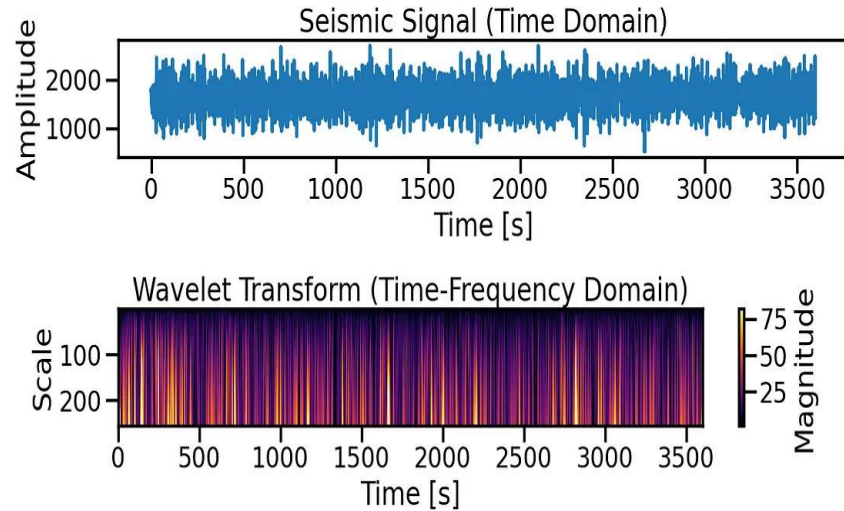


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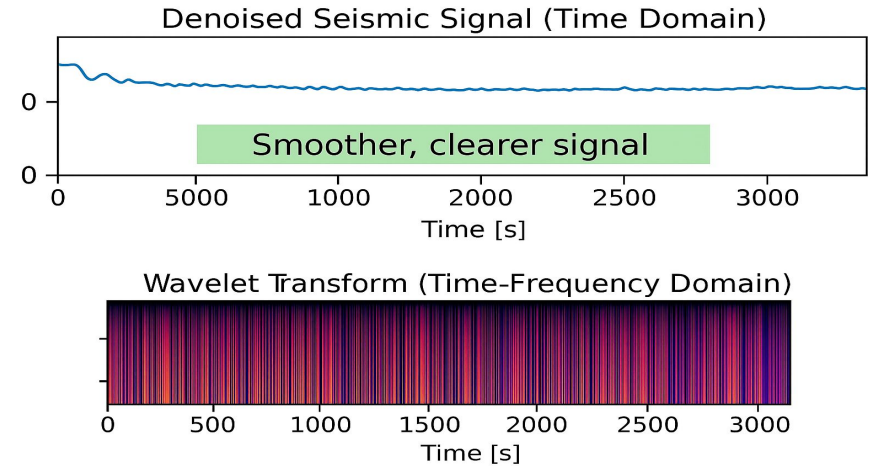


fig: 6 [3]

Impact of the Wavelet Denoising

1. Time-Domain Signal

Before denoising:

- Appears dense and noisy, with many rapid amplitude spikes
- Hard to distinguish actual seismic events from random vibrations

After denoising:

- The waveform becomes smoother
- High-frequency is reduced
- Real seismic activity becomes more visible

Impact of the Wavelet Denoising

2. Wavelet Transform

Before denoising:

- Bright vertical lines at low scales
- These lines represent short bursts of high-frequency noise

After denoising:

- Much of the energy at small scales (top) disappears
- The plot becomes cleaner, especially in the low scale region
- Only the more significant, structured components (e.g., mid-to-high scale seismic activity) remain

References

1. fig :1 <https://ch.mathworks.com/help/wavelet/ug/wavelet-denoising-and-nonparametric-function-estimation.html>
2. Sundling, C Matthew & Sukumar, N. & Zhang, H. & Embrechts, M. & Breneman, Curt. (2006). Wavelets in chemistry and cheminformatics. Reviews in Computational Chemistry. 22. 295-329.
3. practical examples are available at fig:4,5,6
<https://medium.com/pythoneers/wavelet-transform-a-practical-approach-to-time-frequency-analysis-662bdadeb08b>

Thank you very much for your attention!